

CSE200: Online-1 on LaTeX (B2)

Instructions

- Reproduce the following article using $\text{\LaTeX}$ exactly as it is.
- You do not need to reproduce this instruction page.
- Ensure that all text formatting, lists, tables, equations, figures, references are implemented as presented in the article with the appropriate LaTeX commands.

Mark Distribution

Component	Marks
Text Formatting	15
Lists	15
Equations	25
Tables	25
Figures	20
Total	100

CSE200: Online-1 on LaTeX

Your Name (Student ID)

January 5, 2026

1 Introduction

Technical reporting requires **clarity** and *precision* to ensure that information is communicated effectively. In this article, we demonstrate the integration of **text**, *tables*, equations, and **figures** to present structured and meaningful content.

Proper use of **headings**, *lists*, and highlighted text helps organize ideas logically. Additionally, elements such as font variations, inline symbols (x_i, y_j), and controlled spacing contribute to a professional and consistent document layout.

In summary, a well-prepared technical document combines structured sections, accurate notation, and visual components to improve both readability and presentation quality.

2 Key Concepts

2.1 Mixed and Nested Lists

1. Data Acquisition

- Raw data collection
- Sensor-based measurements

2. Data Processing

- z:** Noise removal

- y:** Feature extraction
- x:** Data normalization

3. Result Analysis

- (a) Statistical evaluation
- (b) Model validation
 - Mean error calculation
 - Standard deviation analysis

3 Mathematical Expressions

Mathematical representation is crucial. Consider these examples:

$$E = mc^2 + \frac{1}{2}kx^2 \tag{1}$$

$$\begin{aligned} f(x) &= \sin(x) + \cos(x) \\ &= \sqrt{2} \sin\left(x + \frac{\pi}{4}\right) \end{aligned} \tag{2}$$

$$I = \int_0^\infty e^{-t} dt = 1$$

Piecewise function:

$$h(t) = \begin{cases} t^2 & \text{if } t < 0 \\ \sqrt{t} & \text{if } t \geq 0 \end{cases}$$

$$\begin{aligned} y(x) &= \frac{1}{1+x^2} + \sum_{n=1}^{\infty} \frac{x^n}{n!} \\ &\quad - \ln\left(1 + \frac{x^3}{3!}\right) \end{aligned} \tag{3}$$

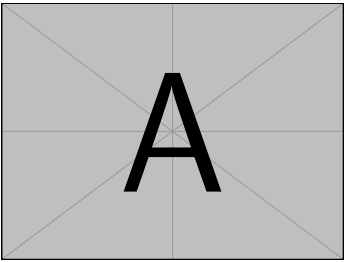
4 Tabular Representation

Experiment	Results		Remarks
	Trial 1	Trial 2	
Test A	78%	82%	Acceptable
Test B	91%	93%	Excellent
	89%	90%	
Test C	45%	50%	Needs Improvement

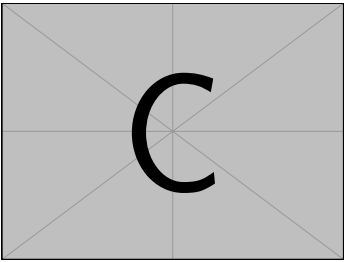
Table 1: Experimental Results

5 Figures

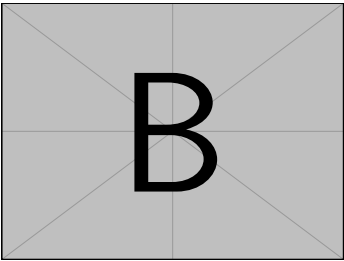
5.1 Two-Row Subfigure Layout



(a) Plot A



(b) Plot C



(c) Plot B

Figure 1: Two-row arrangement: A and C on top, B centered below

Figure 1 demonstrates side-by-side comparison of multiple visual outputs.

6 Conclusion

This document demonstrates **proper structuring** using L^AT_EX with *new content*, equations, tables, and figures. Students should pay close attention to consistency, **alignment**, and *formatting* to achieve full marks.